NIFTY 50 Financial Data Analysis and Forecasting Project Report

1. Project Overview

This project analyzes historical stock market data from the NIFTY 50 dataset to understand stock price movements, trading volume, daily returns, and market volatility.

The analysis uses Python for data cleaning, exploratory data analysis, feature engineering, correlation analysis, and time-series forecasting. ARIMA (5,1,0) was applied to historical RELIANCE closing prices to explore stock price forecasting.

The project also uses PostgreSQL to store and analyze the processed financial data through SQL queries and Power BI to create interactive dashboards.

The main objective is to identify stock performance patterns, examine financial trends, evaluate forecasting performance, and present meaningful insights through data visualization.

2. Dataset Summary

- **Dataset Name:** NIFTY 50 Historical Stock Market Data
- **Date Range:** 1 January 2000 – 30 April 2021
- **Combined Dataset Size:** 235,192 rows and 16 original columns before feature engineering.
- **Key Features:**
    - **Stock Information:** Stock symbol and Date.
    - **Price Information:** Prev Close, Open, High, Low, Last, Close, and VWAP.
    - **Trading Information:** Volume, Turnover, and Trades.
    - **Delivery Information:** Deliverable Volume and %Deliverble.
    - **Engineered Features:** Year, Month, Daily_Return_%, Price_Change, Volatility_20D_%, and Year_Month.
- **Duplicate Records:** Duplicate records were checked and removed during data cleaning.
- **Missing Data:** Missing values in essential fields were handled. Missing values in optional trading fields were reviewed.
- **Final Combined Dataset:** The processed dataset was used for stock-wise analysis and further forecasting.

3. Exploratory Data Analysis Using Python

The dataset was processed and analyzed using Python, Pandas, NumPy, Matplotlib, and Seaborn.

- **Data Loading:** Imported the NIFTY 50 ZIP dataset and extracted individual stock CSV files.
- **Data Combination:** Combined the individual stock files into one DataFrame.
- **Initial Exploration:** Used df.shape, df.head(), df.info(), and df.describe() to understand the dataset structure and numerical statistics.

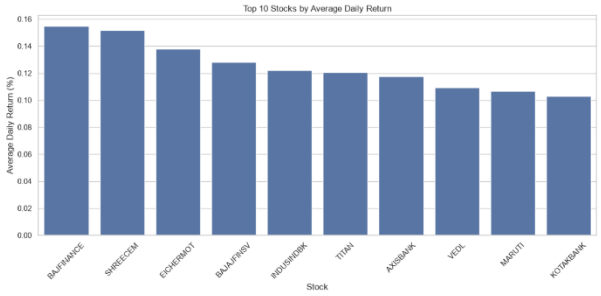
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Dataset shape: (235192, 16)
```

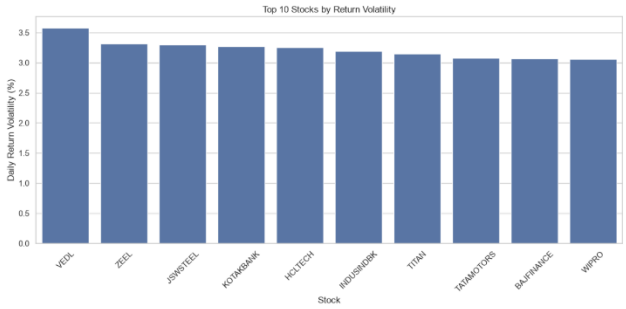
	Date	Symbol	Series	Prev Close	Open	High	Low	Last	Close	VWAP	Volume	Turnover	Trades	Deliverable Volume	%Deliverble	Stock
0	2007-11-27	MUNDRAPORT	EQ	440.0	770.0	1050.0	770.0	959.0	962.9	984.72	27294366	2687719053785000.0	NaN	9859619	0.3612	ADANIPORTS
1	2007-11-28	MUNDRAPORT	EQ	962.9	984.0	990.0	874.0	885.0	893.9	941.38	4581338	431276530165000.0	NaN	1453278	0.3172	ADANIPORTS
2	2007-11-29	MUNDRAPORT	EQ	893.9	909.0	914.75	841.0	887.0	884.2	888.09	5124121	455065846265000.0	NaN	1069678	0.2088	ADANIPORTS
3	2007-11-30	MUNDRAPORT	EQ	884.2	890.0	958.0	890.0	929.0	921.55	929.17	4609762	428325662830000.0	NaN	1260913	0.2735	ADANIPORTS
4	2007-12-03	MUNDRAPORT	EQ	921.55	939.75	995.0	922.0	980.0	969.3	965.65	2977470	287519974300000.0	NaN	816123	0.2741	ADANIPORTS

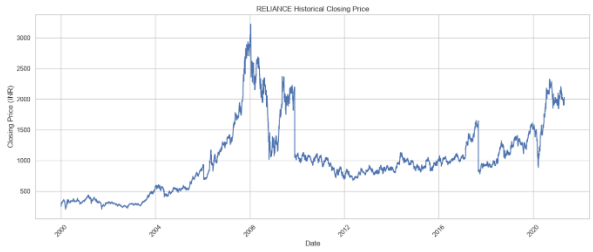
- **Data Cleaning:** Converted Date into datetime format, converted financial columns into numeric data types, checked duplicates, and handled missing values.
- **Data Validation:** Removed records with missing essential fields and invalid closing prices.
- **Feature Engineering:** Created Year, Month, Daily_Return_%, Price_Change, Volatility_20D_%, and Year_Month.
- **Stock Performance Analysis:** Compared stocks based on average closing prices.
- **Daily Return Analysis:** Examined average daily returns across different stocks.
- **Volatility Analysis:** Identified stocks with higher and lower return volatility.
- **Historical Price Analysis:** Examined RELIANCE historical closing prices.
- **Trading Volume Analysis:** Analyzed RELIANCE trading volume over time.
- **Monthly Analysis:** Examined monthly average closing prices, monthly returns, and trading volume.
- **Distribution Analysis:** Visualized daily returns and trading volume distributions.
- **Relationship Analysis:** Examined the relationship between trading volume and daily returns.

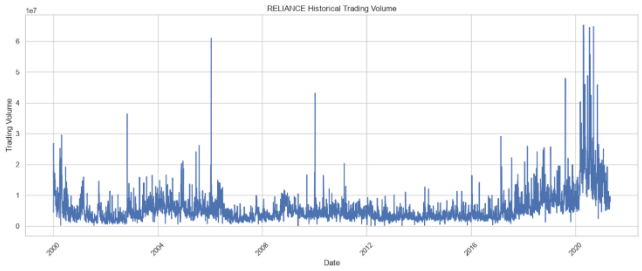
- **Rolling Volatility:** Calculated and visualized 20-day rolling volatility.
- **Correlation Analysis:** Used correlation coefficients and a heatmap to examine relationships between numerical financial features.

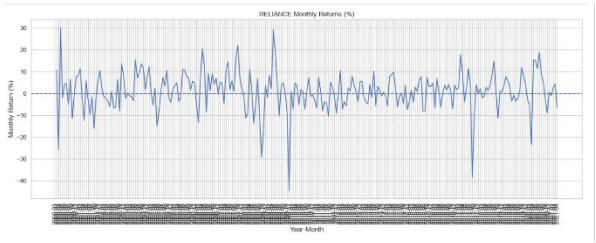


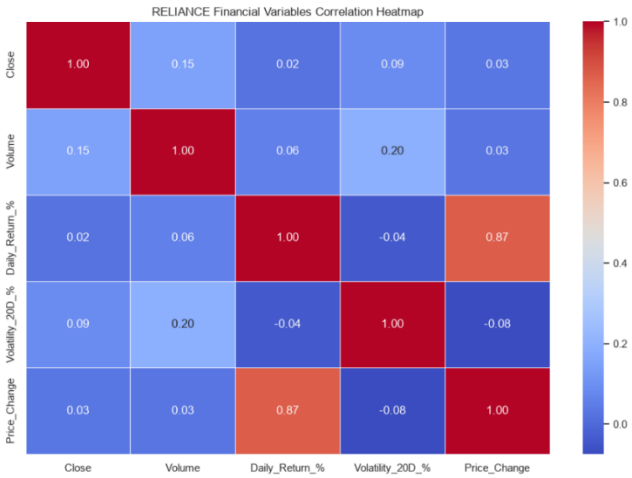












4. Financial Forecasting Using ARIMA

Time-series forecasting was performed using the historical closing prices of RELIANCE.

4.1 Data Preparation

- Selected RELIANCE stock data from the combined financial dataset.
- Sorted the data according to Date.
- Prepared the closing price series for time-series analysis.
- Used an 80:20 chronological split for training and testing.
- Training dataset: 4,244 observations.
- Testing dataset: 1,062 observations.
- Training period: 3 January 2000 – 12 January 2017.
- Testing period: 13 January 2017 – 30 April 2021.

4.2 ARIMA Model Development

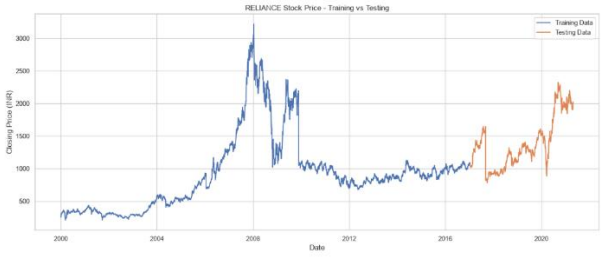
- Applied the ARIMA time-series forecasting algorithm.
- Selected ARIMA model parameters as (5,1,0).
- Trained the model using the training dataset.
- Generated predictions for the testing period.
- Compared predicted closing prices with actual closing prices.
- Visualized actual and predicted stock prices to understand forecasting performance.

4.3 Forecasting Results

The model was evaluated using the following metrics:

- **Mean Absolute Error (MAE):** 365.95
- **Mean Squared Error (MSE):** 239,320.84
- **Root Mean Squared Error (RMSE):** 489.20
- **Mean Absolute Percentage Error (MAPE):** 23.03%

The evaluation indicates that the ARIMA model has noticeable prediction errors. Its predictions were relatively stable across the long testing period, so the model has limitations in capturing changing stock price movements.





SARIMAX Results

Dep. Variable:	Close	No. Observations:	4244
Model:	ARIMA(5, 1, 0)	Log Likelihood	-20727.372
Date:	Fri, 30 Oct 2026	AIC	41466.744
Time:	19:06:05	BIC	41504.862
Sample:	0 - 4244	HQIC	41480.215
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
ar.L1	0.0358	0.008	4.609	0.000	0.021	0.051
ar.L2	-0.0041	0.009	-0.432	0.666	-0.023	0.014
ar.L3	-0.0815	0.009	-9.451	0.000	-0.098	-0.065
ar.L4	-0.0282	0.008	-3.462	0.001	-0.044	-0.012
ar.L5	-0.0417	0.009	-4.751	0.000	-0.059	-0.024
sigma2	1025.7548	2.072	495.086	0.000	1021.694	1029.816

Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	24502853.44
Prob(Q):	0.97	Prob(JB):	0.00
Heteroskedasticity (H):	2.84	Skew:	-10.15
Prob(H) (two-sided):	0.00	Kurtosis:	374.73

5. PostgreSQL Database and SQL Analysis

PostgreSQL was used to store the cleaned financial dataset and analysis outputs for structured querying.

- **Database Name:** nifty50_analysis
- **Database Integration:** Connected Python with PostgreSQL using SQLAlchemy and psycopg2.
- **Data Storage:** Exported processed financial data and analysis results into PostgreSQL tables.
- **Main Tables:**
    - nifty50_cleaned
    - stock_summary
    - reliance_monthly
    - reliance_monthly_volume
    - reliance_correlation
    - reliance_arima_forecast

The following SQL analyses were prepared:

1. Top 10 stocks by average closing price.

2. Highest and lowest average daily returns.
3. Top 10 stocks by return volatility.
4. RELIANCE historical closing price analysis.
5. Monthly trading volume analysis.
6. Monthly return analysis.
7. Trading volume and daily return correlation.
8. RELIANCE 20-day rolling volatility analysis.
9. RELIANCE financial variables correlation analysis.
10. ARIMA forecast performance evaluation.

Power BI was used to create an interactive dashboard using the financial data stored in PostgreSQL.

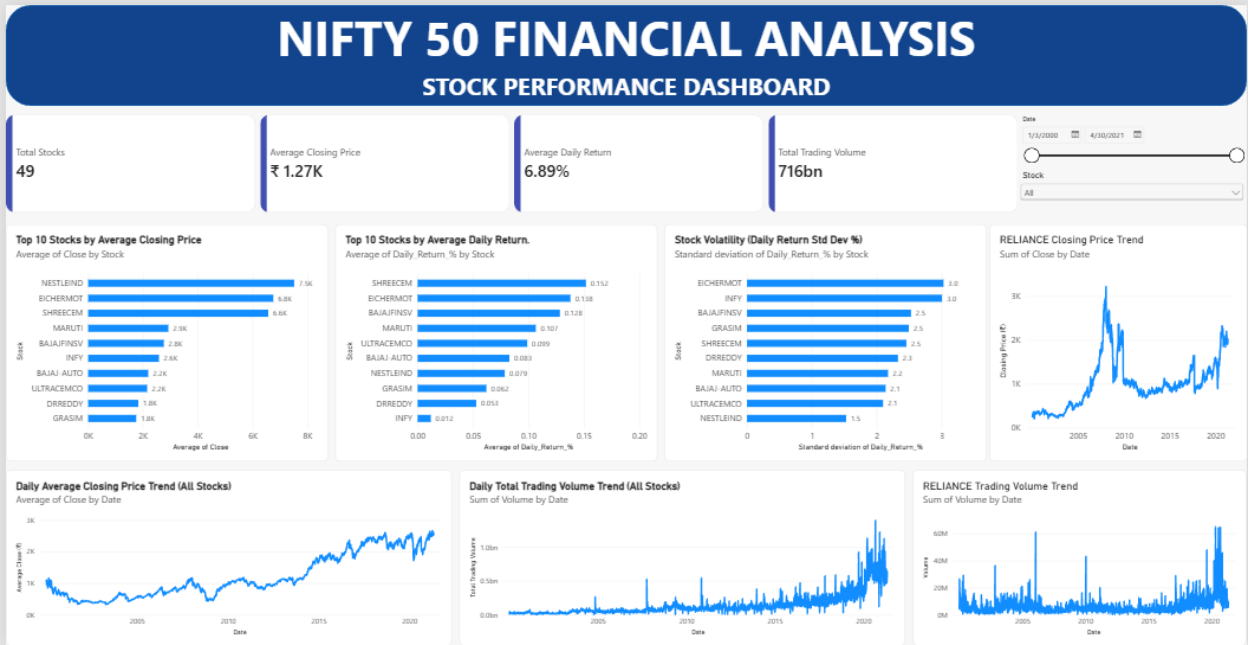
The dashboard was organized into two pages.

6.1 Page 1: NIFTY 50 Stock Performance

The first dashboard page presents an overview of stock performance.

- **Key Performance Indicators:**
    - Total Stocks
    - Average Closing Price
    - Average Daily Return
    - Total Trading Volume
- **Visualizations:**
    - Top 10 stocks by average closing price.
    - Top 10 stocks by average daily return.
    - Stock-wise return volatility.
    - RELIANCE historical closing price trend.
    - RELIANCE trading volume trend.
    - Daily average closing price across the available stocks.
    - Daily total trading volume.

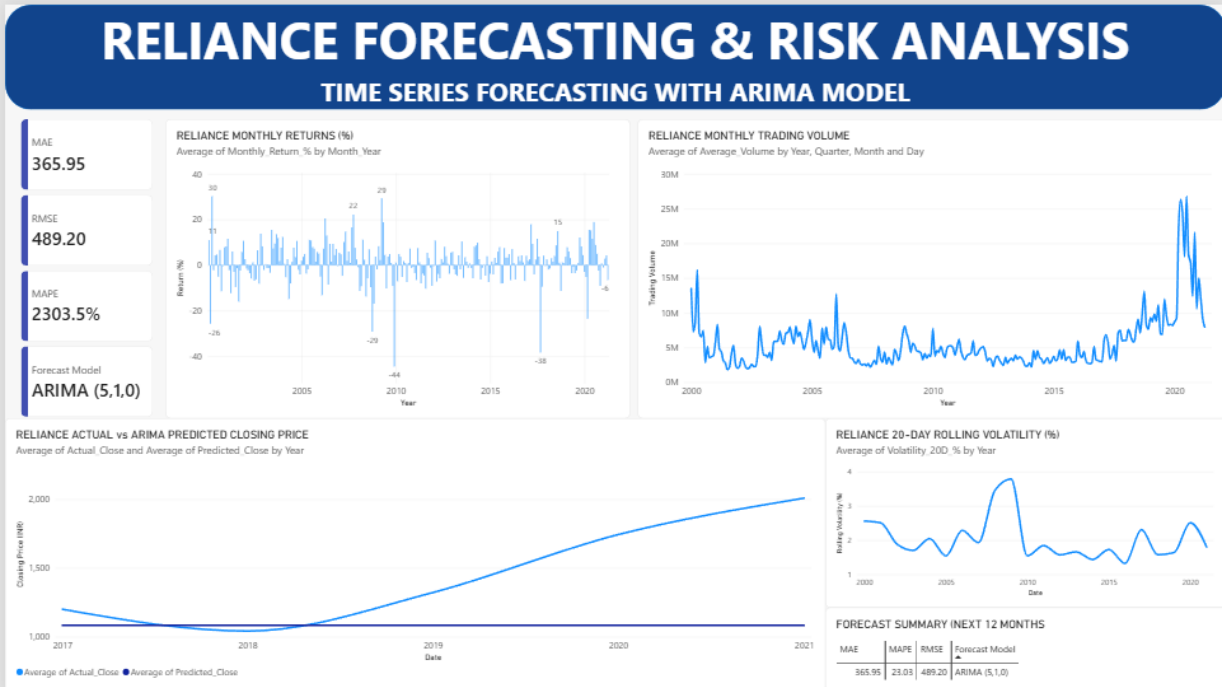
- **Interactive Filters:** Stock and Date slicers.



6.2 Page 2: RELIANCE Forecasting and Risk Analysis

The second dashboard page focuses on RELIANCE historical performance, volatility, and ARIMA forecasting.

- **Key Performance Indicators:**
    - MAE
    - RMSE
    - MAPE
    - Forecast Model
- **Visualizations:**
    - RELIANCE monthly returns.
    - Monthly trading volume.
    - 20-day rolling volatility.
    - Actual versus predicted closing prices.
    - Forecast performance summary.



6.3 Business and Financial Insights

The analysis provides several areas for financial investigation:

- **Stock Performance:** Compare stocks based on average closing prices and daily returns.
- **Return Analysis:** Identify stocks with higher and lower average daily returns.
- **Volatility:** Examine stocks with higher return volatility to understand historical price fluctuations.
- **Trading Activity:** Analyze trading volume patterns to understand changes in market activity.
- **Monthly Trends:** Compare monthly price movements and trading volume.
- **Correlation:** Examine relationships between trading volume, daily returns, and other numerical financial variables.
- **Forecasting:** Use ARIMA evaluation results to understand the limitations of historical stock price forecasting.

Important note: Average closing price alone does not measure investment performance. Historical returns and volatility also need to be considered. The ARIMA model results are based on historical testing data and should not be treated as a prediction of current or future market prices.

7. Tools and Technologies Used

- **Python:** Data cleaning, feature engineering, exploratory analysis, and forecasting.
- **Pandas:** Data manipulation, preprocessing, and aggregation.
- **NumPy:** Numerical operations and data preparation.
- **Matplotlib:** Financial data visualization.
- **Seaborn:** Statistical plots and correlation heatmaps.
- **Statsmodels:** ARIMA time-series forecasting.
- **Scikit-learn:** Forecast evaluation metrics.
- **PostgreSQL:** Storing and querying financial data.
- **SQLAlchemy:** Connecting Python with PostgreSQL.
- **SQL:** Financial data analysis and business queries.
- **Power BI:** Interactive financial dashboards and data visualization.

8. Project Conclusion

The NIFTY 50 Financial Data Analysis and Forecasting project demonstrates an end-to-end financial data analytics workflow involving data extraction, data cleaning, feature engineering, exploratory data analysis, correlation analysis, time-series forecasting, SQL analysis, and dashboard development.

Python was used to combine and process historical stock market data, analyze stock performance, calculate daily returns, and measure rolling volatility. The ARIMA (5,1,0) model was applied to RELIANCE closing prices using a chronological training and testing split.

The forecasting model achieved an MAE of 365.95, RMSE of 489.20, and MAPE of 23.03%. These results indicate that the model has limitations in accurately capturing stock price movements over the testing period.

PostgreSQL was used to store and query the processed financial data, while Power BI was used to present stock performance, trading activity, volatility, and forecasting results through interactive dashboards.

Overall, this project helped demonstrate practical skills in Python, SQL, PostgreSQL, time-series analysis, and Power BI. The findings can support historical financial analysis, but they should not be considered investment advice or guaranteed predictions of future stock prices.

9. Project Deliverables

- Python Jupyter Notebook containing data processing, EDA, and ARIMA forecasting.
- Cleaned NIFTY 50 financial dataset in CSV format.

- Stock-wise summary CSV file.
- RELIANCE monthly analysis CSV files.
- RELIANCE correlation analysis CSV file.
- RELIANCE ARIMA forecast results CSV file.
- PostgreSQL database tables and SQL analysis queries.
- Power BI dashboard with two pages.
- Final project report.